



Title	Phoenix WinNonlin NCA
Document type	Standard Operating Procedure (3.0)
Document ID	SOP-62
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Approval date	Sep 30 2025
Effective date	Sep 30 2025
Next periodic review date	Sep 30 2027

1. Objective

- 1.1. This SOP outlines detailed, step-by-step instructions for the consistent and accurate generation of pharmacokinetic (PK) parameters derived from plasma, urine, and feces using Non-Compartmental Analysis (NCA) with the Phoenix WinNonlin software.

2. Scope

- 2.1. This SOP describes procedures for verifying the accuracy of PK data reported by CROs.
- 2.2. It is not intended for generating PK data for regulatory filings.
- 2.3. The SOP applies to personnel with working knowledge of NCA using Phoenix WinNonlin software.

3. Responsibilities

3.1. PK Analyst/Scientist

- 3.1.1. Performs PK analysis according to this SOP, reviews all outputs, and documents any deviations from the CRO PK analysis.
- 3.1.2. Said deviations are communicated to the CRO and are resolved through a documented review process, with final approval required from both Cybin and the CRO to confirm that the issue has been satisfactorily addressed.

3.2. Quality Check

- 3.2.1. The review process is conducted as a quality check for verification purposes only.

4. Terms and Definitions

Terms	Definitions
μg	Microgram
$A_{\text{ef}}(t_1-t_2)$	Amount of unchanged drug excreted into feces within time span from t_1 to t_2 .
$A_{\text{eu}}(t_1-t_2)$	Amount of unchanged drug excreted into urine within time span from t_1 to t_2 .
$AUC_{(0-\text{inf})}$	The area under the concentration versus time curve from time zero to infinity.
$AUC_{(0-t)}$	The area under the concentration versus time curve from

	the start of dose administration to the last observed quantifiable concentration, using the linear trapezoidal method.
BLOQ	Below the Lower Limit of Quantitation
Cl	Total plasma, serum or blood clearance of drug after intravenous administration.
Cl/F	Apparent total plasma or serum clearance of drug after oral administration
Cl _F	Fecal clearance
Cl _R	Renal clearance
C _{max}	The maximum observed concentration measured after dosing.
CROs	Contract Research Organizations
fe _f	Fraction of administered drug excreted unchanged in feces
fe _u	Fraction of administered drug excreted unchanged in urine
h	hour (s)
IM	Intramuscular
IP	Intraperitoneal
IV	Intravenous
kg	Kilogram
lambda z	First-order rate constant associated with the terminal elimination phase
LLOQ	Lower Limit of Quantitation
mg	Milligram
min	Minute(s)
mL	Milliliter
NCA	Non-Compartmental Analysis
ng	Nanogram
PK	Pharmacokinetics
R ²	Adjusted R ² ; goodness of fit must be ≥ 0.8
SC	Subcutaneous
t _{1/2}	The apparent terminal elimination half life.
t _{max}	The time after dosing at which the maximum

	concentration was observed.
V_z	Volume of distribution during terminal phase after intravenous administration
V_z/F	Apparent volume of distribution during terminal phase after oral / extravascular administration.

5. Procedure

5.1. Data Import and Initial Preparation

5.1.1. Project Creation: Create a new project in Phoenix WinNonlin named with a clear, unique identifier (e.g., "StudyX_PK_Analysis").

5.1.1.1. Import Data: **Import** raw concentration-time data files (e.g., .xlsx, .csv) into the project's Data folder.

5.1.1.2. Map Data: In the **File Import Wizard**, ensure column headers are correctly mapped, and select the Has units row option, if applicable.

5.1.1.3. Alternatively, concentration-time data may be entered by hand by creating a work sheet by mapping column headings (e.g. time, concentration, subject)

5.1.2. After data import, **Send** the data set to **Plotting** (XY plot) to ensure data entry is correct.

5.1.3. Data Preparation

5.1.3.1. LLOQ/BLOQ Handling: Define rules for handling values below the LLOQ. BLOQ values before the first quantifiable concentration are set to zero for NCA. BLOQ values after the first quantifiable concentration are handled as missing.

5.1.3.2. Urine Data Preparation: Ensure urine concentration data is paired with the corresponding volume and urine collection intervals (start and end times).

5.2. Plasma PK Parameter Generation using NCA

5.2.1. Create NCA Object: Right-click on the Workflow in the Object Browser and select Insert > Non-Compartmental Analysis > NCA.

5.2.2. In the NCA **Options** tab:

5.2.2.1. Select Model type: plasma (200-202) or urine (210-212) depending on matrix

5.2.2.2. Select Dose Options: extravascular for oral, SC, IP, IM, or IV bolus, or IV infusion

5.2.2.3. Select dose parameters: (Unit [mg], and Normalization [kg])

5.2.2.4. Select Sparse sampling if applicable

5.2.2.5. Select Calculation Method: Linear up/Log down for the calculation of AUC with Weighting set to Uniform

5.2.3. In the NCA **Rules** tab:

5.2.3.1. Lambda Z Calculation: Set rules for terminal phase estimation where adjusted $R^2 \geq 0.8$ and at least 3 data points of the terminal phase excluding $C_{\max}$

5.2.3.2. AUC Extrapolation to infinity: Set criteria for acceptable AUC extrapolation where %Extrap $\leq 20\%$.

5.2.4. Define **Mappings**: In the Main Mappings panel:

5.2.4.1. Map Subject or ID to the Sort context.

5.2.4.2. Map Time (min or h) to the Time context.

5.2.4.3. Map Concentration ($\mu\text{g/mL}$ or ng/mL) to the Concentration context.

5.2.5. **Specify Dosing**:

5.2.5.1. Single-Dose: In the Dosing tab, use either internal worksheets or link an external dosing file. Map Dose, Time, and Route contexts.

5.2.5.2. Multiple-Dose: For multiple-dose studies, check the Additional Dose Information box and specify the Intervals and Additional Doses.

5.2.6. **Execute and Review**:

5.2.6.1. Click the Execute icon to run the analysis.

5.2.6.2. Review the generated time-concentration plots for each subject to ensure proper terminal phase selection and identify potential outliers.

5.2.6.3. Review the Final Parameters table for any flags or warnings

5.2.6.4. PK parameters mapped in Parameter Names

5.2.6.5. The following PK parameters will be calculated but not limited to: $C_{\max}$, $t_{\max}$, AUC_{0-t} , $\text{AUC}_{0-\text{inf}}$, λ_z , $t_{1/2}$, Cl , Cl/F and V_z , V_z/F

5.2.6.6. No value of λ_z , $\text{AUC}_{0-\text{inf}}$ or $t_{1/2}$ will be reported for cases that do not exhibit a terminal log linear phase in the concentration versus time profile

5.3. Urine/Feces PK Parameter Generation using NCA

5.3.1. Create NCA Object: Insert a separate NCA object for the urine or feces analysis

5.3.2. Define Mappings:

5.3.2.1. Map Subject or ID to the Sort context.

5.3.2.2. Map Start Time and End Time to their respective contexts.

5.3.2.3. Map Concentration and Volume to their respective contexts.

5.3.2.4. Specify Dosing: Link the dosing information as described for plasma analysis.

5.3.2.5. Select Model Type: In the Options tab, set the Model Type to Urine (210-212).

Use Urine (210-212) for feces

5.3.3. Calculate

5.3.3.1. Urine: A_{eu} , fe_u , Cl_R

5.3.3.2. Feces: A_{ef} , fe_f , Cl_F

5.3.4. Execute and Review:

5.3.4.1. Execute the object

5.3.4.2. Review the output, including parameters like A_{e_u} or A_{e_f} or (cumulative amount excreted in urine or feces, respectively) and Cl_R or Cl_F (renal or feces clearance, respectively)

5.4. Reporting and Documentation

5.4.1. Export Results: Export the Final Parameters tables, concentration data, and generated plots into a clear, report-ready format (e.g., .xlsx, .pdf).

5.4.2. QC Check: Conduct a quality control check to verify parameter values and plot interpretations.

5.4.3. Save Project: Save the Phoenix project (.phxproj) with the final workflow, including all analysis steps.

5.4.4. Archival: Archive the project and final output in a designated secure location

6. References

1. Certara Phoenix WinNonlin User Guide.

7. Related Documents

[SOP-63 Reporting of PK Parameters and Descriptive Statistics](#)

8. Attachments